

ADITYA KAMATH

College Park, MD • aditya.kamath.1011@gmail.com • (240) 791-6951 • [LinkedIn](#) • [GitHub](#) • [Portfolio](#)
(Open to Relocate)

Machine Learning Engineer with 4+ years of industry experience leading the design and deployment of enterprise-grade **Generative AI** systems. Specialized in fine-tuning large **LLM** models, building **RAG**-powered assistants, and integrating AI microservices using **LangChain**, **OpenAI**, and **AWS**. Proven success delivering multimodal assistants, intelligent chatbots, and automation platforms that boost efficiency by **70%**. Adept at full-stack AI development with **FastAPI**, **React**, and **Docker**. **Actively seeking opportunities** to drive innovation in **Generative AI**, intelligent automation, and large-scale AI product development.

AWARDS, RECOGNITION

- First Place, AI in Business Case Competition, University of Maryland, 2024
- Graduate Student Government Representative, University of Maryland, 2024–2025
- Featured on the Smith Podcast, University of Maryland, for contributions as an AI Engineer and developing an innovative product for the university, 2025

PROFESSIONAL EXPERIENCE

UNIVERSITY OF MARYLAND

Machine Learning Engineer

College Park, US

Jan 2024 - Sep 2025

Client: University of Maryland - Robert H. Smith School of Business

Project 1: Developed and deployed an AI-powered learning assistant automating MBA assignment grading and mock interviews, saving 100+ hours per semester and improving accuracy; scaled to 8+ faculty, 5 career coaches, and 500+ students via Canvas ELMS integration

- Collaborated with the Dean of Experiential Learning to develop an in-house multimodal, **agentic AI** learning assistant using LLMs with **TTS and STT (AssemblyAI)** capabilities for simulating case, behavioral, technical interviews, and open-ended assignments across **8+ courses**
- Engineered and deployed a **Python**-based multimodal agentic AI assistant integrated with Canvas ELMS, automating grading and interview evaluation for 500+ students—saving over 500 faculty hours each semester
- Designed **SQL-backed FastAPI** microservices to store and retrieve student and assignment data from Amazon RDS, maintaining latency under 3 seconds and supporting real-time updates
- Designed and deployed an AI-powered multimodal learning assistant using **PyTorch**, automating MBA assignment grading and interview evaluations—saving 500+ faculty hours/semester and improving grading accuracy by **70%**
- Designed and implemented robust **MLOps** pipelines, streamlining the deployment, monitoring, and scalability of machine learning models in production environments
- Developed and fine-tuned **Generative AI** models to create high-quality text, images, and audio, improving content generation efficiency by 35% and reducing production costs by 20%
- Architected and deployed a custom **LLM-powered** learning assistant integrating **OpenAI SDK** and **LangChain** to automate assignment grading and mock interviews for **500+ students**, reducing manual effort by 70%
- Implemented fine-tuning and **Retrieval-Augmented Generation (RAG)** pipelines, integrating structured and unstructured data sources to improve model accuracy and contextual relevance
- Integrated **Model Context Protocol (MCP)** to enable consistent context sharing between AI agents, ensuring smoother coordination and modularity across GenAI systems
- Designed and implemented an **ETL pipeline** to collect and process public data (articles, YouTube videos) using **Docling**, creating fine-tuning datasets and enhancing them via model distillation with reasoning models
- Optimized **multi-agent** reasoning chains for concurrency, latency, and cost efficiency, maintaining **p95 latency <3s** while reducing prompt calls per evaluation by **40%**
- Developed a **React-based** frontend with interactive dashboards and a **FastAPI backend** integrating **AWS S3** (file storage) and **Amazon RDS** (data management), enhancing usability and data accessibility

- Led faculty and career coach adoption, saving **500+ manual evaluation** hours per semester and improving grading consistency via **AI agents** implementing course- and coach-specific rules

Client: Office of the Comptroller of Maryland

Project 2: Developed an AI-powered Employee Learning Management System prototype for Maryland's 21CFSE initiative, featuring a dashboard and RAG-based chatbot to assist employees with queries, business analysis, and scaling strategy

- Applied insights from the business process analysis to inform the design and architecture of the AI-enhanced platform, ensuring alignment with identified workflows and directly addressing bottlenecks through targeted AI integrations for optimized report retrieval and document handling.
- Developed a generative AI-powered website for the Comptroller of Maryland's 21CFSE program using **AWS**, integrating Amazon Bedrock and Amazon Lex for an **RAG chatbot** to convert natural language queries into **SQL**, enabling efficient retrieval of reports.
- Integrated guardrails into the AI system, including content filters to prevent the generation of harmful or biased outputs, role-based access controls for secure data handling, and audit logging to monitor system usage and maintain accountability for government operations.
- Built a React-based frontend and **Flask** backend with RESTful APIs, using Amazon DynamoDB for fast FAQ and document searches and S3 for scalable, cost-efficient storage, **reducing document access time by 40%**.
- Conducted extensive research on **AWS services**, cost optimization strategies, and data privacy regulations to ensure low operational costs, compliance with government security standards, and robust data protection measures throughout the platform's development

WEBACY (EXTERN)

AI Engineer

Remote, USA

Jun 2024 - Aug 2024

Developed a Python-based AI agent for Webacy to analyze smart contract audit data scraped from multiple sources. The system classified vulnerabilities, identified high-risk patterns, and helped prioritize auditing efforts based on prevalence and severity

- Leveraged a search **API (SerpAPI)** to discover and retrieve smart contract audit reports across various categories, ensuring a diverse and comprehensive dataset for analysis.
- Utilized Firecrawl to scrape and retrieve smart contract audit reports from multiple reputable auditing websites, compiling a comprehensive dataset of **500+** audits to uncover vulnerability patterns and support Webacy's security assessments.
- Employed **LangChain** and OpenAI's models to set up AI agents for processing and classifying audit data into Webacy's proprietary risk tag scheme, enabling efficient categorization of vulnerabilities such as reentrancy and overflow errors.
- Conducted in-depth statistical analysis and trend reporting using **Python (Pandas, NumPy, Matplotlib)** and visualized results, identifying the most prevalent and co-occurring vulnerabilities to inform audit strategies and **improve detection efficiency by 25%** for client projects

CAPGEMINI

Software Engineer

Mumbai, India

Nov 2020 - Oct 2022

Client: A top 5 globally, US-based medical device company

Developed an AI-powered Employee Learning Management System prototype for Maryland's 21CFSE initiative, featuring a dashboard and RAG-based chatbot assisting employees with queries, business analysis, and planning for AI-driven ELMS scaling

Project 1: Automation and testing for pacemaker programmer software

- Developed and maintained scripts to automate testing for pacemaker software and programming interfaces on **Azure Cloud**, ensuring robust functionality and reducing manual testing time by **40%** through streamlined test execution.
- Conducted data analysis on **clinical** trial datasets (50,000+ records) using Python (Pandas, NumPy) and Excel, identifying trends in pacemaker performance metrics to support optimization, contributing to a 20% improvement in battery efficiency, while ensuring HIPAA-compliant handling of PII.
- Implemented traceability matrices and documentation processes to map test cases to **FDA** regulatory requirements, ensuring comprehensive coverage and facilitating audit readiness for pacemaker software validations.

- Collaborated with cross-functional teams in **agile Scrum** meetings to validate test results and share insights via Excel dashboards, enhancing team decision-making and **accelerating development cycles by 25%**.

Project 2: Development of features for pacemaker programmer software during firmware and hardware upgrade from legacy to modern systems.

- Designed and developed backend software for pacemaker programmer interfaces, implementing C++ based APIs to enable secure data exchange between the programmer device and cloud-based remote monitoring systems, achieving a 50% reduction in data transfer delays while meeting **FDA** and **HIPAA** standards.
- Built and tested server-side logic for the programmer application, ensuring reliable data processing and integration with clinician-facing interfaces, supporting accurate device configuration and patient data management.
- Wrote comprehensive unit and integration test suites using **C++** to validate backend functionality, addressing issues and ensuring system reliability, contributing to successful regulatory audits through clear documentation.
- Applied knowledge from **ISO 13485:2016** and **ISO 14971:2019** training courses to ensure the programmer's software met medical device quality and risk management standards, **supporting a 25% improvement in device efficiency** via optimized data handling.
- Collaborated with teams using **Agile** methodologies and **Jira** for task tracking, participating in design reviews, and maintaining audit-ready documentation, keeping projects on track and aligned with industry standards

PROJECTS

REDDIT CONSENSUS

Created a multi-agent tool for analyzing and condensing discussions from online forums

- Built Reddit Consensus, a multi-agent app using FastAPI, OpenAI Agents SDK, Tavily Search API, and PRAW (Python wrapper for Reddit's API) to aggregate and summarize Reddit discussions with source-verified consensus.
- Leveraged Matplotlib for server-side chart generation and React with Axios for a dynamic frontend, supporting concurrent query rewriting and post-analysis

SPEAKERLENS

Built an analysis platform for extracting insights from audio conversations

- Developed an intelligent audio analysis tool that performs speaker diarization, transcription, and visualization of conversation dynamics using **AssemblyAI** and OpenAI APIs, enabling actionable insights from multi-speaker audio recordings in meetings, sales calls, and research interviews.
- Built an interactive Streamlit dashboard with Matplotlib and Plotly, integrating a custom RAG pipeline with reranking using LangChain and ChromaDB for context-aware QA, sentiment analysis, entity detection, and real-time speaker analytics

FINE-TUNING LANGUAGE MODELS

Adapted AI models for specialized language processing applications

- Fine-tuned a range of LLMs and SLMs using Hugging Face Transformers and Unsloth for domain-specific NLP tasks, including grammar correction, text summarization, and sentiment analysis, with optimization via LoRA and QLoRA.
- Conducted local inference using Ollama and deployed models on AWS SageMaker in the cloud

SKILLS

- **Programming Languages:** Python (Pandas, NumPy, Matplotlib, LangChain, Unsloth), SQL, JavaScript
- **Tools & Libraries:** AWS (Bedrock, Lex, S3, RDS, SageMaker, KnowledgeBase), OpenAI, AssemblyAI, Hugging Face Transformers, Unsloth, FastAPI, React, Flask, Streamlit, Plotly, Docling, Azure Cloud, Excel, Git, Docker
- **Machine Learning:** LLMs, Fine-tuning (LoRA, QLoRA), RAG Pipelines, Agentic Workflows, Model Distillation, NLP (Sentiment Analysis, Entity Detection)
- **Databases:** Amazon DynamoDB, Amazon RDS
- **Other:** Agile Scrum, Web Scraping (SerpAPI, Firecrawl), Data Visualization, TTS/STT

EDUCATION

UNIVERSITY OF MARYLAND, ROBERT H. SMITH SCHOOL OF BUSINESS <i>MS in Information Systems</i>	College Park, US Aug 2023 - Dec 2024
MUMBAI UNIVERSITY <i>B.E. in Electronics Engineering</i>	Mumbai, India Nov 2020